# PROTOCOL ELEMENTS DOCUMENT

## TITLE PAGE

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| **Full Title:** | **A Phase 1 Study of JNJ-99526687, an Actinium-225-Labeled Antibody Targeting Delta-Like Ligand 3 (DLL3), for Small Cell Lung Cancer** |
| **Sponsor Protocol Identifier:** | 99526687SCC1001 |
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| **Sponsor's Investigational Product Codes:** | JNJ-99526687<br>JNJ-99526726 |
| **Trial Phase:** | Phase 1 |
| **Sponsor Name and Address:** | Janssen Research & Development LLC, 920 Route 202 South, Raritan, 08869 New Jersey, USA |

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# 1. SCHEDULE OF ACTIVITIES

**Table 1 Schedule of Activities for All Dose Schedules**

| **Assessments/Procedures** (may be modified by SET)<br>**P1 = Part 1 only**<br>**S = Selected cohorts only** | **Screening** (≤30 days prior to Imaging) | **Imaging** (7-10 days prior to dosing D1) | **Dose 1 and Subsequent Doses** (Visit window ±2 days) | | | | | | | | **EOT** | **Follow-Up** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | **Day -7[d]** | **Day 1** | **Day 8** | **Day 15** | **Day 29** | **Day 43** | **Day 50** | **Q4W until next dose** | | |
| ICF, demographics, medical hx, disease characteristics, height | X | | | | | | | | | | | |
| Weight | X | S | | | | | | | | | | |
| Eligibility or retreatment criteria[a] | X[a] | S[a] | X[a] | X[a] | | | | | | | | |
| **Safety assessments** (Assessments done ≤72 hrs prior to the dose do not need to be repeated.) | | | | | | | | | | | | |
| Vital signs (temperature, HR, RR, BP) | X | | | Pre and 1h post | P1 | X | | | | X | | |
| $SpO_2$ (resting and ambulatory) | X | | X | X | P1 | X | X | X | X | X | | |
| Physical examination and ECOG | X | | X | X | | | | | | X | | |
| Hematology and chemistry | X | S | X | X | P1 | X | X | X | X | X | | |
| Serology and coagulation | X | | | | | | | | | | | |
| Urinalysis | X | | | | | X[b] | | X[b] | | | | |
| Pregnancy testing | X[c] | S[c] | | X[c] | | | | | | | | |
| 12-lead triplicate ECG | | | | Pre and 1h post | | | | | | | | |
| ECHO or MUGA scan | X | | | | | | | | | | | |
| Pulmonary function testing[e] | X[e] | | | | | | | | X[e] | X[e] | | |
| **Study drug administration** | | | | | | | | | | | | |
| JNJ-99526726 (Zr89) | | S | | | | | | | | | | |
| JNJ-99526687 (Ac225) | | | | X | | | | | | | | |
| **Imaging/Disease assessments (**Baseline disease assessments may be performed within 42 days prior to Dose 1.) | | | | | | | | | | | | |
| PET for Zr89 | | Selected – initial participants will be serially imaged. SET to confirm optimal timepoint for Zr89 imaging | | | | | | | | | | |
| SPECT for Ac225 | | | Selected – timing determined by the SET | | | | | | | | | |
| CT chest/abdomen/pelvis/brain (or MRI) | X | | Every 6 (±1) weeks after first dose of JNJ-99526687 for first 24 weeks then every 12 (±1) weeks. | | | | | | | | | |
| PK, PD and immunogenicity | Refer to Table 2 | | | | | | | | | | | |
| **Monitoring** | | | | | | | | | | | | |
| Adverse events | Continuous from signing of ICF up to 30 days after last dose of study drug or until the start of subsequent anticancer therapy (if earlier). | | | | | | | | | | | |
| Concomitant therapy | | | | | | | | | | | | |
| Subsequent anticancer treatment and survival (can be via phone call) | | | | | | | | | | | Q12W | |

a. For first dose, participants must meet eligibility criteria when ordering study drug (Day -7 ± 1 day) and on day of study drug administration. For subsequent doses, participants must meet retreatment criteria when ordering study drug (Day -7 ± 1 day) and on day of study drug administration. The pre-dose laboratory assessment can be performed 1 working day prior to study drug administration. Laboratory parameters are listed in Appendix 7.2.

b. If Grade ≥1 proteinuria is observed, monitor at each site visit along with blood pressure or perform 24-hour urine collection for further evaluation.

c. For participants of childbearing potential only, a serum or urine pregnancy test must be conducted during screening and within 72 hours of the first dose of study treatment and each subsequent dose.

d. Can be combined with Imaging visit if on same day.

e. Pulmonary function tests at screening should include spirometry, corrected DLCO, and lung volumes. At subsequent timepoints, PFTs will include spirometry and DLCO (with lung volumes if clinically indicated). May be obtained up to 4 weeks after Day 50 provided they are obtained prior to any subsequent study drug dose.

**Table 2: Schedule of Activities for Pharmacokinetic, Pharmacodynamic, and Immunogenicity Assessments for Part 1 and Part 2**

| Dose | Day | Time point [a] | Window | PK (antibody) | PK (radioactivity) | Immuno-genicity | Archived Tissue [b] | Fresh Biopsy [c] |
|---|---|---|---|---|---|---|---|---|
| **Sample** | | | | **Serum** | **Serum** | **Serum** | **Tissue** | **Tissue** |
| **Screening** | | | | | | | X | Optional |
| **Dose 1** | **Day 1** | Predose | ≤4 hours | X | X | X | | |
| | | 5 min | +10 min | X | X | | | |
| | | 1 hour | ±15 min | X | X | | | |
| | | 4 hours | ±35 min | X | X | | | |
| | **Day 3** | 48 hours | ±4 hours | X[e] | X | | | |
| | **Day 8** | | ±1 day | Part 1 | Part 1 | | | |
| | **Day 15** | | ±2 days | X | X | X | | |
| | **Day 29** | | ±2 days | X | X | X | | |
| | **Day 43** | | ±2 days | X | X | X | | Optional |
| **Dose 2** | **Day 1** | Pre-dose | ≤4 hours | X | | X | | |
| | | 5 min | +10 min | X | | | | |
| | **Day 15** | | ±2 days | X | | | | |
| **All subsequent doses** | **Day 1** | Pre-dose | ≤4 hours | Dose 3 and 4 | | Dose 3 and 4 | | |
| **EOT Visit** | | | | X | X | X | | |

The frequency of the pharmacokinetic/immunogenicity assessments may be revised based on review of available data by the SET.

a. Except for pre-dose timepoints, all other timepoints are counted from end of injection of the study drug.

b. Availability of archival tissue (slides or blocks) should be confirmed during screening but not shipped to the sponsor-designated laboratory until after dosing. If available, tumor samples collected within 15 months prior to providing informed consent will be requested.

c. Optional fresh biopsy collections for selected cohorts will begin as determined by the SET.

# 2. INTRODUCTION

## 2.1. Purpose of the Trial

This is a first-in-human, open-label, dose escalation study evaluating the safety, tolerability, pharmacokinetics, dosimetry, and preliminary anti-tumor activity of JNJ-99526687 in SCLC. The study employs an integrated theranostic design in which pre-treatment Zr89 PET imaging with the companion imaging agent JNJ-99526726 will be performed.

SCLC is a highly aggressive form of lung cancer with a very poor prognosis. Current SoC brings the 5-year survival to only 9%, dropping to 4% for patients with ES disease, highlighting a clear unmet need. DLL3 is a notch ligand of particular interest in SCLC due to its high tumor selectivity, with little expression in normal adult tissues but high prevalence in tumor. This target has also been clinically validated with tarlatamab-dlle (IMDELLTRA® [Amgen]), a TCE-targeting DLL3 receiving US FDA approval in 2024 for treatment of second-line ES-SCLC. While tarlatamab shows a moderately improved duration of response in this highly pretreated population, the objective response rate remains relatively low, demonstrating the need for alternative approaches that are potentially less reliant on immune cell infiltration.

**JNJ-99526687** is a ~100 kDa bivalent spFv-Fc targeting DLL3, with stochastic conjugation to the 6,6'-((1,4,10,13-tetraoxa-7,16-diazacyclooctadecane-7,16-diyl)bis(methylene))dipicolinic acid chelator JNJ-95981860 and radiolabeled with Ac225. The mechanism of cytotoxicity is through DLL3 binding of JNJ-99526687 followed by internalization; Ac225 decay releases alpha particles that induce double-strand DNA breaks. In vivo, JNJ-99526687 demonstrated an optimal biodistribution profile showing reduced circulation exposure relative to a standard mAb and with sustained accumulation in the tumor and minimal normal tissue exposure, and potent efficacy against SCLC tumors.

**JNJ-99526726** is the same bivalent spFv-Fc with stochastic conjugation to the deferoxamine chelator and radiolabeled with Zr89. JNJ-99526726 will be used for imaging.

For the most comprehensive nonclinical and clinical information regarding JNJ-99526687, refer to the latest version of the IB for JNJ-99526687 once available.

# 3. TRIAL OBJECTIVES AND ASSOCIATED ESTIMANDS

## 3.1. Primary Objectives

| Primary Objectives | Endpoints |
|---|---|
| • Determine the RP2D regimens of JNJ-99526687 | • Incidence and severity of adverse events including DLT per NCI-CTCAE criteria v5.0 based on clinical assessments |
| • Determine the optimal mass dose of JNJ-99526726 | • PET imaging and serum PK assessments, including clearance and tissue accumulation |
| • Determine the safety and tolerability of JNJ-99526687 at the RP2D regimens in selected disease cohorts | • Incidence and severity of all adverse events at RP2D regimens |

## 3.2. Secondary Objectives

| Secondary Objectives | Endpoints |
|---|---|
| • Preliminary antitumor activity of JNJ-99526687 | • Objective response rate, time to response and duration of response according to RECIST v1.1.<br>• PFS |
| • To evaluate the PK profiles of JNJ-99526687, total antibody, payload and major metabolite | • PK parameters including $C_{max}$, $C_{trough,}$ $T_{max}$, $AUC_{0-\tau}$, and accumulation ratio |
| • To assess the immunogenicity of JNJ-99526687 | • Incidence of ADAs against JNJ-99526687 |
| • Determine the safety of JNJ-99526726 | • Incidence and severity of adverse events per NCI-CTCAE criteria v5.0 based on clinical assessments |
| • Determine the ability of JNJ-99526726 to act as an imaging agent for DLL3 | • Tissue and tumor distribution on PET imaging |

## 3.3. Exploratory Objectives

| Exploratory Objectives |
|---|
| • To explore the relationships between PK, adverse event profile, and clinical activity |
| • To investigate biomarkers predictive of clinical response or resistance to JNJ-99526687 including but not limited to the expression of DLL3 from tumor samples |

# 4. TRIAL DESIGN

## 4.1. Description of Trial Design

This is a FIH, Phase 1 study to evaluate the safety, PK, PD, and preliminary antitumor activity of JNJ-99526687 administered to adult participants with ES-SCLC. Additional DLL3 tumor types may be considered based on imaging and emerging clinical data. Approximately 100 participants will be treated in this 2-part study (60 participants in Dose Escalation and 40 participants in Dose Expansion). A SET consisting of all participating investigators and key sponsor personnel, chaired by the Sponsor Study Responsible Physician, will be commissioned for this trial.

JNJ-99526726 (Zr89 PET imaging agent) will be administered intravenously at 2 mCi at the assigned antibody mass dose, followed by PET imaging to assess biodistribution, normal tissue dosimetry, and tumor targeting.

- The starting antibody mass dose will be 3 mg, with escalation up to 20 mg permitted based on SET review of biodistribution data to ensure linear pharmacokinetic behavior without target saturation or non-specific sink effects.
- Initial participants will undergo serial PET imaging at multiple timepoints (0-4h, 24±6h, 42-78h, and 120-216h post-administration) to determine the optimal imaging window for subsequent participants.
- If it is determined that the biodistribution is unfavorable (e.g. normal tissue uptake is significantly greater than tumor uptake), the participant will be offered repeat imaging with a new administration of JNJ-99526726 at a revised mass dose before a determination is made regarding eligibility to proceed to therapeutic dosing.
- Once sufficient biodistribution and dosimetry data have been established, the SET may waive the pre-treatment imaging requirement for specific cohorts.

JNJ-99526687 (therapeutic Ac225) will be administered intravenously approximately 7-10 days after the imaging agent at an assigned radioactivity dose (μCi/kg) and at a mass dose determined as optimal by the imaging agent.

- Dosing with JNJ-99526687 will start at 0.75 μCi/kg (based on non-clinical cynomolgus monkey studies) on 3 mg antibody.
- Radioactivity dose escalation is governed by the BOIN design, based on the rate of DLTs observed within a 42-day DLT observation window. SET dose escalation decisions will be based on all available data including safety, PK, dosimetry, and preliminary anti-tumor activity.
- Subsequent doses may be given at least 8 weeks after the previous dose in the absence of disease progression or toxicities until the cumulative absorbed dose to any organ is projected to exceed EBRT-equivalent organ thresholds based on available dosimetry data. SET may consider decreasing dosage for subsequent doses for safety or to account for decreased tumor burden.

Once the provisional RP2D is determined, the study will move to Dose Expansion.

**Efficacy Evaluations**

Clinical activity will be assessed using brain, chest, abdomen, and pelvis CT scans (or MRI as necessary), plus other areas of known disease involvement as appropriate. Response will be evaluated by the investigator according to the RECIST, version 1.1.

**Pharmacokinetic and Immunogenicity Evaluation**

Blood samples will be collected to characterize the pharmacokinetics of serum radioactivity, the concentration of JNJ-99526687, and to characterize the presence of anti-drug antibodies to JNJ-99526687.

### 4.1.1. Safety

Safety will be monitored by the SET at each dose-escalation step and at regular intervals during dose expansion. Cumulative data from subsequent treatment doses will also be monitored and cumulative dose relative to EBRT-equivalent upper limits. SET decisions will be based on the review of all available data including PK, safety, and preliminary anti-tumor activity.

The safety of JNJ-99526687 will be assessed by vital signs, and physical examinations, ECGs, clinical laboratory tests, pregnancy testing, ECOG performance status, and AE monitoring, including DLTs. Concomitant medications will be recorded. The severity of AEs will be assessed using NCI-CTCAE, version 5.0.

To allow adequate safety monitoring at each new dose level, treatment will be staggered by a minimum of 1 week between the first participant enrolled in a cohort and subsequent participants treated at a dose level that has not yet been declared safe by the SET.

### 4.1.2. Post-Treatment Dosimetry – Ac225 SPECT Imaging

Following administration of JNJ-99526687, SPECT imaging will be performed to characterize the in vivo distribution of the Ac225-labeled therapeutic agent and to estimate absorbed dose to normal tissues and tumor. The timing and schedule of Ac225 SPECT imaging will be determined by the SET based on emerging data and will be documented in the imaging charter or a protocol amendment as data mature.

Ac225 SPECT imaging will provide post-treatment dosimetry confirmation and will complement pre-treatment dosimetry estimates derived from Zr89 PET imaging. Absorbed dose estimates from both modalities will inform cumulative dose monitoring relative to EBRT-equivalent organ thresholds.

### 4.1.3. Dose Limiting Toxicity Criteria

A DLT is defined as any of the treatment-related AEs listed in Table 3 that occur during the DLT evaluation period. The DLT evaluation period is defined as the 42-day period after the first dose of JNJ-99526687, based on the expected nadir of bone marrow suppression period.

All adverse events and adverse events fulfilling the DLT criteria will be reviewed and confirmed by the SET. In addition, the cumulative safety profile (all applicable data) may be reviewed by the

SET as part of the continued safety evaluation. Adverse events will be evaluated according to NCI CTCAE Version 5.0.

**Table 3: Dose-limiting Toxicity Criteria [a]**

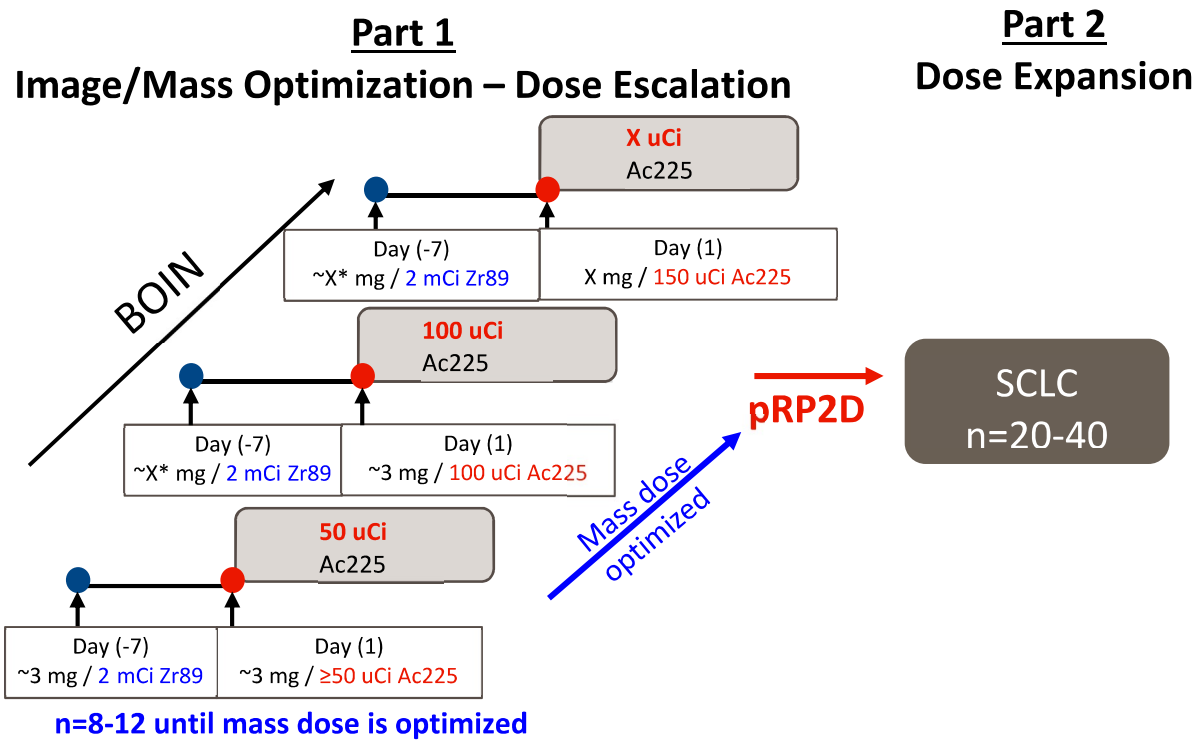
| **Hematologic Toxicity (related to study treatment)** |
|---|
| Grade 4 neutropenia for >5 days or febrile neutropenia |
| Grade 3 thrombocytopenia with bleeding or Grade 4 thrombocytopenia |
| Any Grade 5 |
| **Non-hematological Toxicity (related to study treatment)** |
| Grade ≥2 pneumonitis/ILD |
| Any Grade ≥3 non-hematological toxicity, except for the following:<br>• Grade 3 fatigue, fever, constipation, or diarrhea lasting <7 days with best supportive care<br>• Grade 3 nausea or vomiting lasting ≤48 hours that resolves to Grade ≤1 either spontaneously or with best supportive care<br>• Grade ≥3 ALT or AST that resolves to Grade ≤1 or baseline within 7 days, unless criteria for Hy's law are met [b]<br>• Isolated Grade ≥3 ALP or GGT increase that returns to Grade ≤1 or baseline within 7 days<br>• Grade ≥3 lipase or amylase increase not associated with clinical or radiological evidence of pancreatitis<br>• Grade ≥3 electrolyte abnormalities that last ≤72 hours resolving spontaneously or with best supportive care |

a. Any chemistry abnormalities Grade ≥3 occurring during the DLT period must be reassessed to confirm the grade and resolution to Grade ≤2.

b. Hy's Law criteria defined as ALT or AST value ≥3×ULN, total bilirubin ≥2×ULN, and ALP ≤2×ULN; with no alternative etiology. For participants with baseline Grade 2 elevation of AST or ALT due to liver metastasis, ALT or AST >3× baseline or AST or ALT >8×ULN, whichever is lower, combined with total bilirubin >2×baseline and >2×ULN will be considered meeting Hy's law.

## 4.2. Trial Schema

**Figure 1: Schematic Overview of the Trial**




# 5. TRIAL POPULATION

## 5.1. Description of Trial Population and Rationale

There is an unmet need for improved treatment strategies for recurrent ES-SCLC (see Section 2.1). This trial will include adult participants with recurrent ES-SCLC following progression after platinum-based chemotherapy and immune checkpoint inhibitor.

Screening for eligible participants will be performed within 30 days before administration of trial intervention. The inclusion and exclusion criteria for enrolling participants in this trial are described below. If there is a question about these criteria, the investigator must consult with the appropriate sponsor representative and resolve any issues before enrolling a participant in the trial. Waivers are not allowed.

## 5.2. Inclusion Criteria

Each potential participant must satisfy all of the following criteria to be enrolled in the trial:

| | |
|---|---|
| **Age** | |
| 1. | At the time of informed consent, be ≥18 years of age and at least the legal age of majority in the jurisdiction in which the study is taking place. |
| **Type of Participant and Disease Characteristics** | |
| 2. | Have histologically confirmed ES-SCLC |

<table>
<tr><td>3.</td><td>Have measurable or evaluable disease:<br>Part 1: Either measurable or evaluable disease.<br>Part 2: At least 1 measurable lesion per RECIST v1.1.</td></tr>
<tr><td>4.</td><td>Progressed during or after at least 1 prior therapy including platinum-based chemotherapy and immune checkpoint inhibition, unless participant was ineligible to receive them, and is without therapeutic alternatives.</td></tr>
<tr><td>5.</td><td>Toxicity related to prior anticancer treatment must have resolved to Grade 1 or baseline (except alopecia, vitiligo, peripheral neuropathy or endocrinopathies that are stable on hormone replacement, which must be Grade 2 or better).</td></tr>
<tr><td colspan="2">Clinical Characteristics</td></tr>
<tr><td>6.</td><td>Have an ECOG performance status of 0 or 1.</td></tr>
<tr><td>7.</td><td>Have an eGFR, calculated with the CKD-epi formula using adjusted BSA (at https://www.kidney.org/professionals/gfr_calculator), of ≥50 ml/min.</td></tr>
<tr><td>8.</td><td>Have the following lab values:
<table>
<tr><td></td><td colspan="2">Hepatic metastases</td></tr>
<tr><td></td><td>No known</td><td>Yes</td></tr>
<tr><td>AST</td><td><2.5 x ULN</td><td><4 x ULN</td></tr>
<tr><td>ALT</td><td><2.5 x ULN</td><td><4 x ULN</td></tr>
<tr><td>Total bilirubin</td><td><1.5 x ULN</td><td><3 x ULN</td></tr>
<tr><td>Bilirubin in case of known congenital nonhemolytic hyperbilirubinemias such as Gilbert’s Syndrome</td><td>isolated total bilirubin ≥1.5 x ULN with conjugated [direct] bilirubin <1.5 x ULN</td><td>isolated total bilirubin ≥3 x ULN with conjugated [direct] bilirubin <1.5 x ULN</td></tr>
</table></td></tr>
<tr><td>9.</td><td>Have the following values independent of transfusion within 14 days, or growth factors within 21 days prior to first dose of study treatment:
<table>
<tr><td>Hemoglobin</td><td>≥9 g/dL</td></tr>
<tr><td>Neutrophils</td><td>≥1.5 x $10^9$/L, or<br>≥1 x $10^9$/L if participant has history of benign ethnic neutropenia</td></tr>
<tr><td>Platelets</td><td>≥100 x $10^9$/L</td></tr>
</table></td></tr>
</table>

| | |
|---|---|
| **Sex and Contraceptive/Barrier Requirements** | |
| 10. | While on study intervention and for 6 months after the last dose of study intervention, a participant must:<br>a. Not breastfeed or be pregnant.<br>b. Not donate gametes (ie, eggs or sperm) or freeze for future use for the purposes of assisted reproduction.<br>c. Wear an external condom.<br>d. Practice at least 1 highly effective method of contraception; if oral contraceptives are used, a barrier method of contraception must also be used.<br>e. If of childbearing potential, have a negative highly sensitive (eg, β-hCG) pregnancy test at screening and within 24 hours before the first dose of study treatment, and agree to further pregnancy tests.<br>See Protocol Appendix on Contraceptive and Barrier Guidance for details. |
| **Informed Consent** | |
| 11. | All participants must consent to provide an archived tumor tissue sample (if available) at screening. |
| 12. | Must sign an ICF indicating that they understand the purpose of, and procedures required for, the study and are willing to participate in the study. |
| 13. | Willing and able to adhere to the lifestyle restrictions specified in the protocol. |

## 5.3. Exclusion Criteria

Any potential participant who meets any of the following criteria will be excluded from participating in the trial:

| | |
|---|---|
| **Medical Conditions** | |
| 1. | Participant has a history of clinically significant cardiovascular disease within 6 months prior to signing informed consent including:<br>• Myocardial infarction<br>• Severe or unstable angina<br>• Coronary artery bypass grafting<br>• Clinically significant arrhythmias (e.g., ventricular arrhythmias)<br>• New York Heart Association Class III or IV congestive heart failure<br>• Myocarditis<br>• Cerebrovascular accident or transient ischemic attack<br>• Any grade arterial event |

| | |
|---|---|
| 2. | Venous thromboembolic events (eg, pulmonary embolism or intracardiac thrombus) within 30 days prior to first dose of study treatment; uncomplicated (Grade ≤2 deep venous thrombosis is allowed). |
| 3. | Prior history of ILD, pneumonitis within the prior 12 months prior to enrolment, or other clinically significant pulmonary compromise, particularly a requirement for supplemental oxygen use (>2 L/min by nasal cannula) to maintain adequate oxygenation. |
| 4. | History of solid organ or hematologic stem cell transplantation. |
| 5. | Known history of myelodysplastic syndrome, leukemia, or hematological malignancy with features suggestive of myelodysplastic syndrome/acute myeloid leukemia. |
| 6. | Prolonged QTcF >470 msec.<br>Note: Participants with cardiac pacemakers who are clinically stable are eligible. |
| 7. | Evidence of a clinically significant active viral, bacterial, or fungal infection requiring systemic treatment within 7 days before the first dose of study treatment. |
| 8. | Had major surgery, significant trauma, or has not recovered adequately from surgery within 2 weeks prior to the first administration of study treatment. Must not have major surgery planned during the time the participant is receiving study treatment. |
| 9. | Has a prior or concurrent second malignancy (other than the disease under trial), with the exception of:<br>• Malignancies whose natural history or treatment would likely not interfere with any trial safety endpoints or the efficacy of the trial interventions.<br>• Malignancies that are not progressing nor requiring treatment change in the last 2 years and not considered at high risk of recurrence requiring systemic therapy.<br>• Malignancies treated within the last 2 years and considered at very low risk for recurrence, including:<br>o Non-muscle invasive bladder cancer (solitary Ta-PUNLMP or low-grade, <3 cm, no CIS).<br>o Non-melanoma skin cancers treated with curative therapy or localized melanoma treated with curative surgical resection alone.<br>o Non-invasive cervical cancer.<br>o Breast cancer: adequately treated lobular carcinoma in situ or ductal carcinoma in situ, localized breast cancer and receiving antihormonal agents.<br>o Localized prostate cancer (M0, N0) with a Gleason Score ≤7, treated locally only (RP/RT/focal treatment).<br>o Other malignancies that are considered at minimal risk of recurrence. |
| **Disease Characteristics** | |

| | |
|---|---|
| 10. | Concurrent use of any other anticancer treatment (including non-palliative radiotherapy) or investigational agent; prior anticancer treatments (including non-palliative radiotherapy) must be discontinued for at least 2 weeks or 5 half-lives, whichever is shorter, before the first dose of study treatment, with the following exceptions:<br>a. Prior tarlatamab, immune checkpoint inhibitors, or thoracic radiation must be discontinued at least 4 weeks before starting study treatment.<br>b. Prior CAR-T administration must be at least 90 days prior to starting study treatment |
| 11. | Wide-field radiotherapy (eg, >30% of marrow-bearing bones) within 4 weeks, or focal radiation with palliative intent inside the field of measurable disease within 2 weeks prior to first dose of the study drug. Palliative radiotherapy outside the field of measurable disease during the study is permitted. |
| 12. | Active CNS involvement with the exception of locally treated brain metastases that are clinically stable and asymptomatic for >2 weeks and who are off or receiving low-dose corticosteroid treatment (≤10 mg prednisone or equivalent) for at least 2 weeks prior to start of study treatment. |
| 13. | Participants who are known to be human immunodeficiency virus-positive and meet any of the following:<br>a. Detectable viral load (ie, ≥50 copies/mL) at screening<br>b. CD4+ count ≤300 cells/$mm^3$ at screening<br>c. AIDS-defining opportunistic infection within 6 months of screening<br>d. Not receiving HAART. Any changes in HAART due to resistance/progression should occur at least 3 months prior to screening. A change in HAART due to toxicity is allowed up to 4 weeks prior to screening.<br>**Note**: HAART that could interfere with study treatment is excluded (consult the sponsor for a review of medications prior to enrollment). |
| 14. | Active hepatitis of infectious origin.<br>a. Seropositive for HBV: defined by a positive test for HBsAg. Participants with resolved infection (ie, participants who are HBsAg negative with antibodies to total anti-HBc must be screened using RT-PCR measurement of HBV DNA levels. Those who are RT-PCR positive will be excluded. Participants with serologic findings suggestive of HBV vaccination (anti-HBs positivity as the only serologic marker) AND a known history of prior HBV vaccination, do not need to be tested for HBV DNA by RT-PCR (see Appendix: Hepatitis B Virus Screening).<br>b. Known HCV infection or positive serologic testing for anti-HCV. Participants with positive HCV antibody due to prior resolved disease can be enrolled only if a confirmatory negative HCV RNA test is obtained at screening or within 3 months prior to first dose of study treatment. |

| | |
|---|---|
| | c. Other clinically active liver disease of infectious origin. |
| 15. | Known allergies, hypersensitivity, or intolerance to excipients of JNJ-99526687 or JNJ-99526726 (refer to IBs). |
| 16. | Received or plans to receive any live, attenuated vaccine within 30 days before the first dose of study treatment or within 4 weeks after the last dose of study treatment. Non-live and non-replication-competent vaccines approved or authorized for emergency use (eg, COVID-19) by local health authorities are allowed in accordance with vaccine recommendations or local labeling. |
| 17. | Received immunosuppressive doses of systemic medications, such as glucocorticoids (doses >10 mg/day prednisone or equivalent) within 7 days prior to the first dose of study treatment. A single course of glucocorticoids is permitted as prophylaxis for imaging contrast (ie, for participants with allergies to contrast). |
| **Other Exclusion** | |
| 18. | Any condition for which, in the opinion of the investigator, participation would not be in the best interest of the participant (eg, compromise the well-being) or that could prevent, limit, or confound the protocol-specified assessments. |

**NOTE:** The protocol will describe options for retesting. The required source documentation to support meeting the enrollment criteria will be noted in the protocol.

## 5.4. Lifestyle Considerations

| | |
|---|---|
| 1. | Carry a “wallet study card” with pertinent information about the study for the duration of study participation. |
| 2. | Blood donation is not allowed at any time during the study and at least 100 days after the study following the rules of local blood donation organizations. |
| 3. | Before and after study drug injection, the participant should drink 1 to 2 glasses of water. |
| 4. | Participants are encouraged to observe the following practices for 6 weeks after each study treatment:<br>• Immediately report cough, dyspnea, fever or any worsening of respiratory symptoms.<br>• Stay well hydrated (eg, 6 or more glasses of water per day).<br>• Sit down on the toilet to prevent urine splatter. Flush the toilet twice after each use, and wash hands well after using the toilet each time.<br>• Caregivers must wear disposable gloves if handling urine or soiled clothing from the participant. Dispose of gloves properly in a waste container.<br>• Items such as bedpans or urinals should be thoroughly washed with soap and running water. |

| | |
|---|---|
| | • The participant's laundry must be washed separately from the rest of the household, including towels and sheets. |

# 6. TRIAL INTERVENTION AND CONCOMITANT THERAPY

Trial interventions are all pre-specified, investigational and noninvestigational medicinal products, medical devices and other interventions (eg, surgical and behavioral) intended to be administered to the trial participants during the trial conduct.

## 6.1. Study Treatments Administered

For this study, "study drug" refers to JNJ-99526687 or JNJ-99526726, which are considered investigational medicinal products. Both drug products include labeled and unlabeled spFv-Fc to achieve the desired mass dose.

## 6.2. Retreatment Criteria

Subsequent doses are feasible starting at 8 weeks post treatment for participants meeting the retreatment criteria. This interval may be modified by the SET. Longer intervals between doses may be considered if agreed between the investigator and medical monitor. Any dose adjustment must be overseen by medically qualified study-site personnel (principal or subinvestigator). Retreatment criteria provided in Table 4 must be met 7 (±1) days before the scheduled dose to ensure adequate time to order the study drug.

**Table 4: Study Drug Retreatment Criteria**

| Retreatment Criteria |
|---|
| Must have adequately recovered from toxicity and be off transfusions or growth factors for at least 7 days before the next administration of study drug |
| Hemoglobin ≥9 g/dL |
| Platelets $\geq 100 \times 10^9$/L |
| Absolute neutrophil count $\geq 1.5 \times 10^9$/L |
| Serum total bilirubin <1.5×ULN. Participants with congenital bilirubinemia such as Gilbert's Syndrome may be retreated if conjugated bilirubin is within normal range |
| ALT and AST ≤2.5×ULN (or ≤4×ULN for participants with tumor involvement in the liver) |
| Calculated or measured creatinine clearance >50 mL/min |
| Grade ≤1 or baseline proteinuria |
| No signs or symptoms of respiratory insufficiency |
| Ambulatory $SpO_2$ ≥92%, above baseline, or within 4 percentage points of baseline |
| Chest CT without findings consistent with pneumonitis or ILD unless a clear alternative etiology has been established and the findings have resolved |
| Response per RECIST 1.1 of NonCR/NonPD, stable disease, partial response, or complete response at most recent disease evaluation within 42 days of dose and in the absence of clinical disease progression |

Maximum cumulative dose will be defined in the protocol.

# 7. APPENDIX

## 7.1. APPENDIX 1: GLOSSARY OF TERMS AND ABBREVIATIONS

| abbreviation | description of abbreviated term |
|---|---|
| ADA | anti-drug antibody |
| AE | adverse event |
| ALP | alkaline phosphatase |
| ALT | alanine aminotransferase |
| anti-HBc | hepatitis B core antigen |
| anti-HCV | HCV antibody |
| AST | aspartate aminotransferase |
| $AUC_{0-\tau}$ | area under the drug concentration-time curve from time zero to the end of a specific dosing interval |
| β-hCG | beta-human chorionic gonadotropin |
| BOIN | Bayesian Optimal Interval |
| BP | blood pressure |
| BSA | body surface area |
| CAR-T | chimeric antigen receptor T-cell |
| CIS | carcinoma in situ |
| CKD-epi | chronic kidney disease - epidemiology |
| $C_{max}$ | maximum drug concentration |
| CT | computed tomography |
| $C_{trough}$ | lowest concentration of a drug measured in a patient's bloodstream, occurs immediately before the next scheduled dose is administered |
| DLCO | diffusing capacity of lung for carbon monoxide |
| DLL3 | delta-like ligand 3 |
| DLT | dose-limiting toxicity |
| EBRT | external beam radiation therapy |
| ECG | electrocardiogram |
| ECOG | Eastern Cooperative Oncology Group |
| eGFR | estimated glomerular filtration rate |
| EOT | end of treatment |
| ES | extensive-stage |
| FIH | first in human |
| FOIA | Freedom of Information Act |
| GGT | gamma-glutamyltransferase |
| HAART | highly active antiretroviral therapy |
| HBsAg | hepatitis B surface antigen |
| HBV | hepatitis B virus |
| HCV | hepatitis C virus |
| HIV | human immunodeficiency virus |
| HR | heart rate |
| Hx | history |
| IB | Investigator's Brochure |
| ICF | informed consent form |
| ILD | interstitial lung disease |
| mAb | monoclonal antibody |
| MRI | magnetic resonance imaging |
| NCI-CTCAE | National Cancer Institute Common Terminology Criteria for Adverse Events |
| NonCR | non-complete response |
| NonPD | non-progressive disease |
| P1 | Part 1 |
| PD | pharmacodynamics |
| PET | positron emission tomography |
| PFS | progression-free survival |
| PFT | pulmonary function test |
| PK | pharmacokinetics |

| | |
|---|---|
| pRP2D | punitive recommended Phase 2 dose |
| PUNLMP | papillary urothelial neoplasm of low malignant potential |
| Q#W | every # weeks |
| QTcF | corrected QT interval by Fridericia |
| RESIST | Response Evaluation Criteria in Solid Tumors |
| RP | radical prostatectomy |
| RP2D | recommended phase 2 dose |
| RR | respiratory rate |
| RT | radiation therapy |
| RT-PCR | reverse transcription polymerase chain reaction |
| S | selected |
| SCLC | small cell lung cancer |
| SET | study evaluation team |
| SoC | standard of care |
| SPECT | single-photon emission computed tomography |
| spFv-Fc | single-chain fragment variable - fragment crystallizable |
| $SpO_2$ | peripheral capillary oxygen saturation |
| TCE | T-cell engager |
| $T_{max}$ | time to maximum concentration |
| ULN | upper limit of the normal range |

## 7.2. APPENDIX 2: CLINICAL LABORATORY TESTS

### Protocol-required Safety Laboratory Assessments

| Laboratory Assessments | Parameters | |
|---|---|---|
| Hematology | -hemoglobin<br>-absolute lymphocyte count<br>-platelet count<br>-absolute neutrophil count | |
| Clinical Chemistry | -sodium<br>-potassium<br>-blood urea nitrogen/urea<br>-creatinine<br>-glucose (fasting or random)<br>-aspartate aminotransferase<br>-alanine aminotransferase<br>-gamma-glutamyltransferase | -total bilirubin<br>-direct bilirubin [a]<br>-alkaline phosphatase<br>-phosphate<br>-calcium<br>-albumin<br>-total protein |
| Routine Urinalysis[b] | Dipstick<br>-specific gravity<br>-pH<br>-glucose<br>-protein<br>-blood<br>-ketones<br>-bilirubin<br>-urobilinogen<br>-nitrite<br>-leukocyte esterase | Sediment (if dipstick result is abnormal)<br>-red blood cells<br>-white blood cells<br>-epithelial cells<br>-crystals<br>-casts<br>-bacteria |
| Screening Only Tests | • Coagulation (prothrombin time / international normalized ratio and activated partial thromboplastin time)<br>• Serology [c]: Hepatitis B (HBsAg, anti-HBc, anti-HBs and DNA) and hepatitis C (anti-HCV and HCV-RNA) | |

a. Direct bilirubin if Gilbert's disease.
b. Suggested urine dipstick evaluation; however, perform according to locally available test equipment.
c. See inclusion criteria (Section 5.2) to determine eligibility for enrollment in the study.